

CMS TEAM PROJECT · TU DRESDEN · SOSE 2026

Moderating the emotional tone of AI data stories

What we measured, on 32 runs across eight datasets, judged blind by
Claude Opus 5 against 25 stories our own team wrote from the same data.

Experiments only · every figure generated from the run database

THE RESULT

Moderation lands machine stories on the human level

RAW, ALARMISM

3.74

the generator's own tone

AFTER MODERATION

2.09

a 1.5-point move

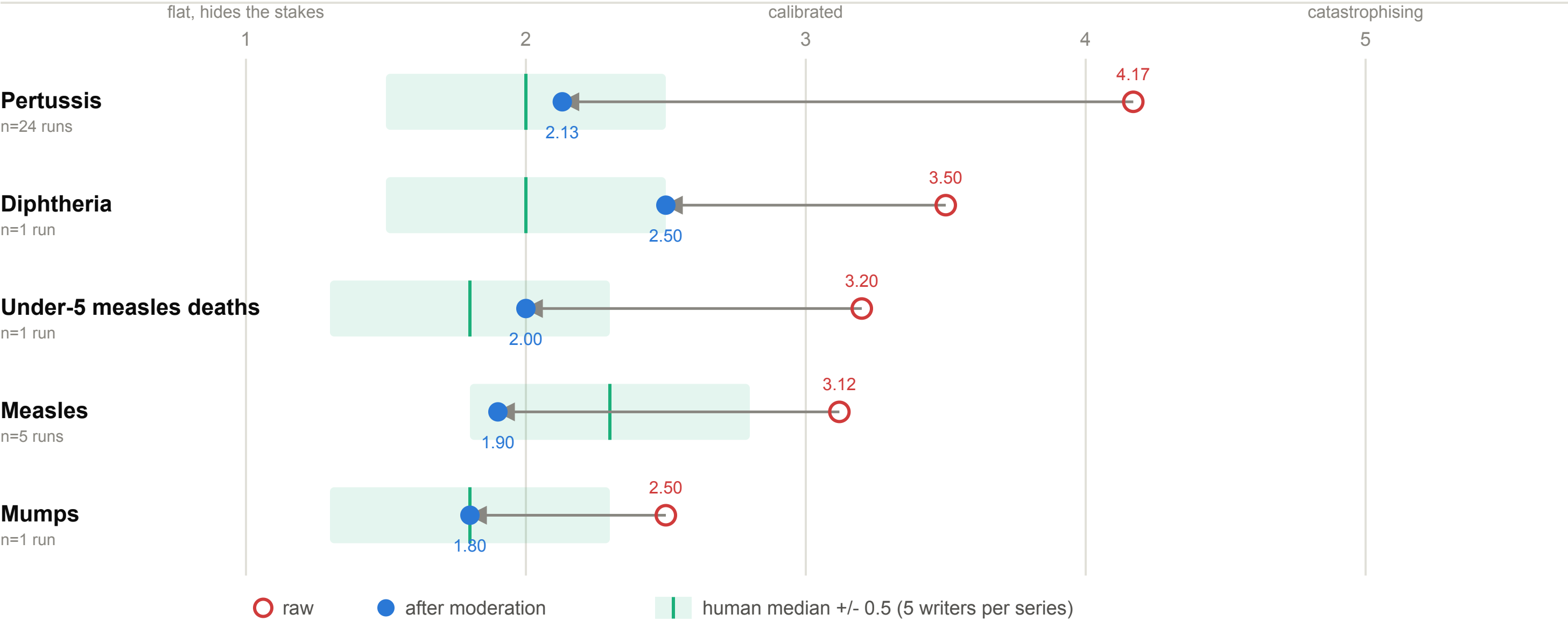
OUR WRITERS

2.00

25 stories, same evidence packs

Moderation moves machine stories onto the human level

Alarmism 1-5, Claude Opus judging blind, one story at a time. Overall 3.74 -> 2.09, human median 2.00.



n=1 per cell on four of five series, so each arrow is one run, not a mean of repeats. The human stories carry no headline and the machine stories do; headlines concentrate alarmism, so the human line is if anything flattered and every gap shown is a lower bound.

WHY OUR METRIC HAS TWO AXES

A falling series fails in the opposite direction

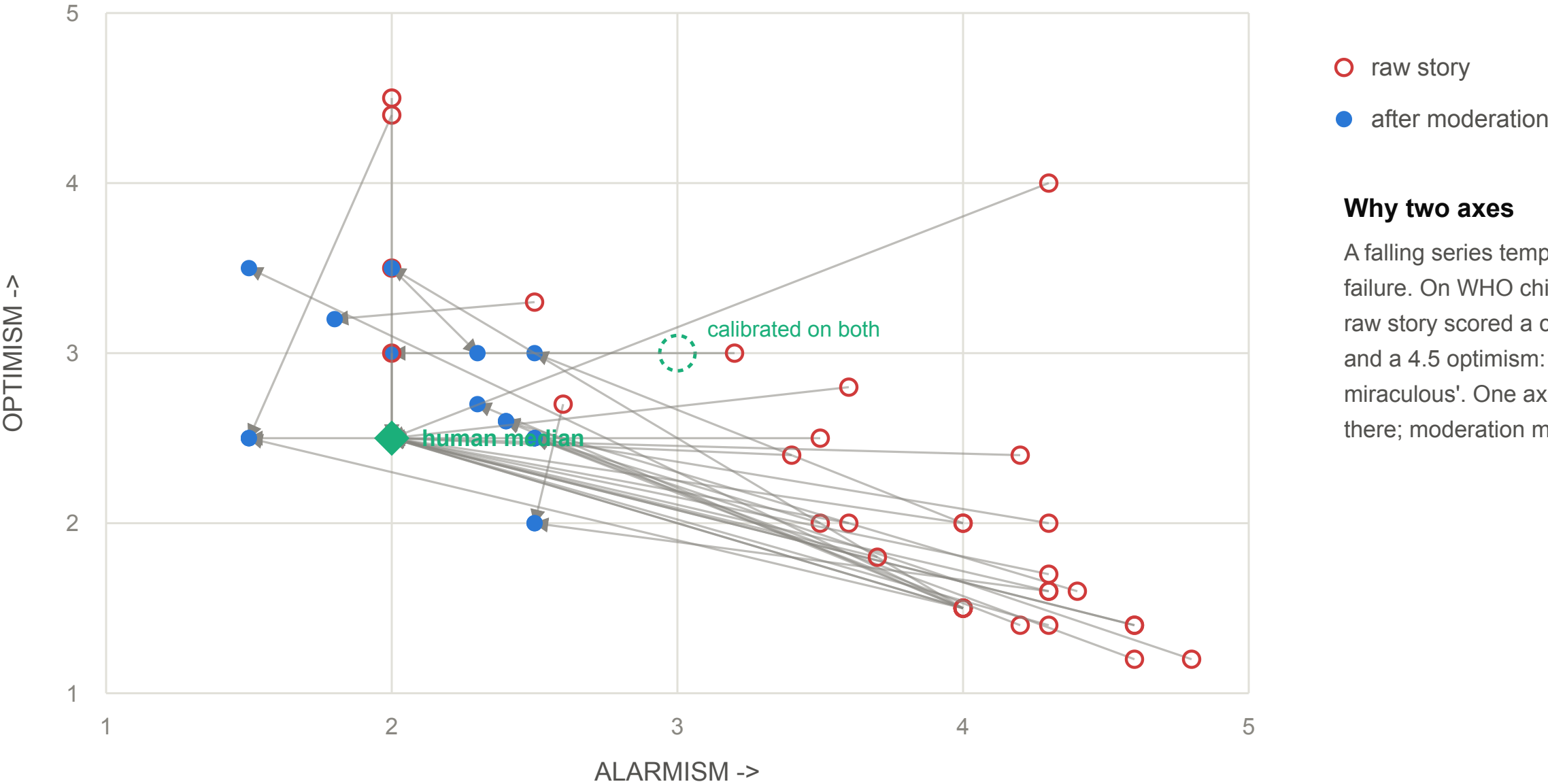
On WHO child mortality the raw story scored a calm **2.0** for alarmism while calling two decades of progress "nothing short of miraculous". Optimism caught it at **4.5**, and moderation moved it to 2.5.

A single-axis judge reports that moderation did nothing on that dataset.

alarmism 3.74 → 2.09 · optimism 2.15 → 2.60

Two axes, because a falling series fails in the other direction

Every run: 35 raw-to-moderated moves, all eight datasets.



Alarmism falls 3.74 -> 2.09 and optimism rises 2.15 -> 2.60, both toward the middle. Note the moderated cloud settles below alarmism 3: the moderator aims at quiet rather than at calibrated, and lands near where these human writers also sit (diamond).

Six combinations tie on tone. Faithfulness breaks the tie.

The moderator converges on the same tone whoever wrote the draft (2.10 vs 2.10, $p = 1.00$). What separates the generators is how many of the raw story's figures survive.

QWEN3.5:4B × GEMMA4:31B

61%

figures kept, ± 20 over 5 seeds

LLAMA3.1:8B × GEMMA4:31B

90%

figures kept, ± 15 over 5 seeds

THE LESSON WE NEARLY PUBLISHED

With one run per cell, our ranking was backwards

A single run had qwen3.5:4b keeping **71%** of figures and llama3.1:8b keeping **50%**, so we recommended qwen3.5:4b.

Five seeds each: **61%** and **91%**. Both original numbers were unrepresentative draws from distributions spanning 33-83% and 67-100%.

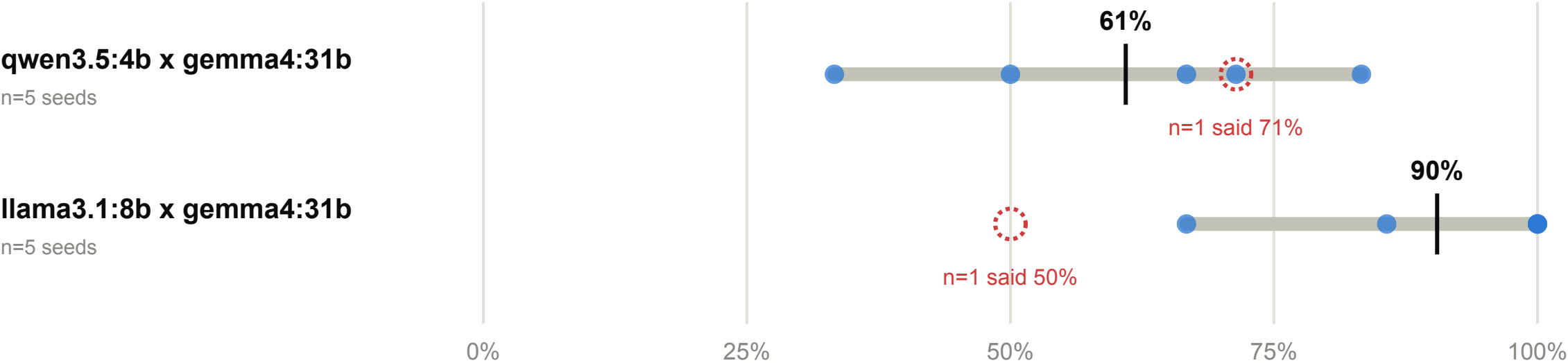
`n = 1 cannot tell a model from a draft`

Judge noise is 0.08 points · run-to-run spread is 0.42 · the variance is the generator's

Five seeds per cell, and the recommendation flips

Same two configurations, run five times each at distinct seeds, pertussis-global.

NUMERIC RETENTION - the share of the raw story's figures that survive moderation

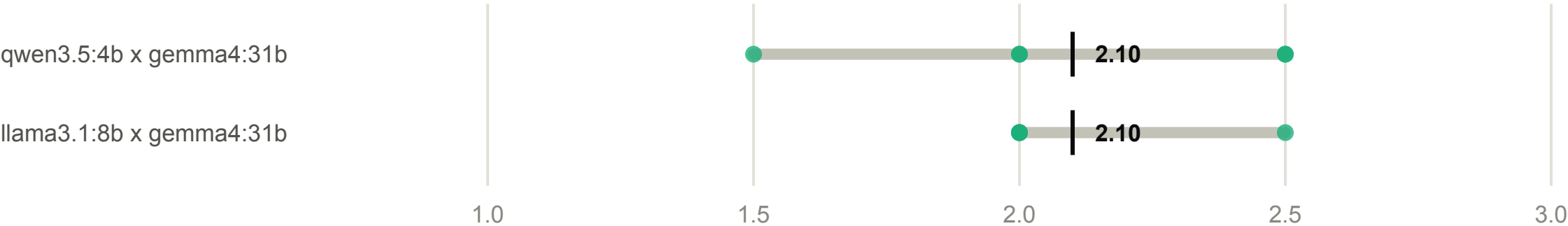


The ranking was **backwards**

One run each had put qwen3.5:4b ahead on retention, 71% to 50%. Five seeds each reverse it: 61% to 91%.

Welch t = -2.70, p = 0.029, Cohen d = 1.71.

MODERATED ALARMISM - where the moderator leaves the story

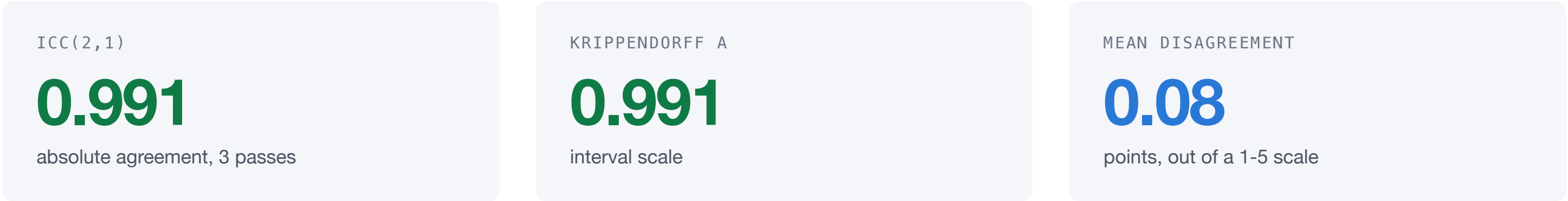


Identical: 2.10 both, p = 1.00.
The moderator converges on the same tone whoever wrote the draft.

Seeds differ deliberately: repeats at a fixed seed replay the same text and measure determinism, not stability. Judge noise is not the source of this spread - three independent passes over the same story agree to ICC 0.99, mean spread 0.08, against a run-to-run alarmism spread of 0.42. The variance is the generator's, not the judge's.

CAN THE INSTRUMENT BE TRUSTED

The tone scale has an error bar



Same model and prompt each pass, so this is self-consistency: an upper bound on what a different judge would agree to.

SCALE OR LINEAGE

Writing calmly is a family trait, not a size effect

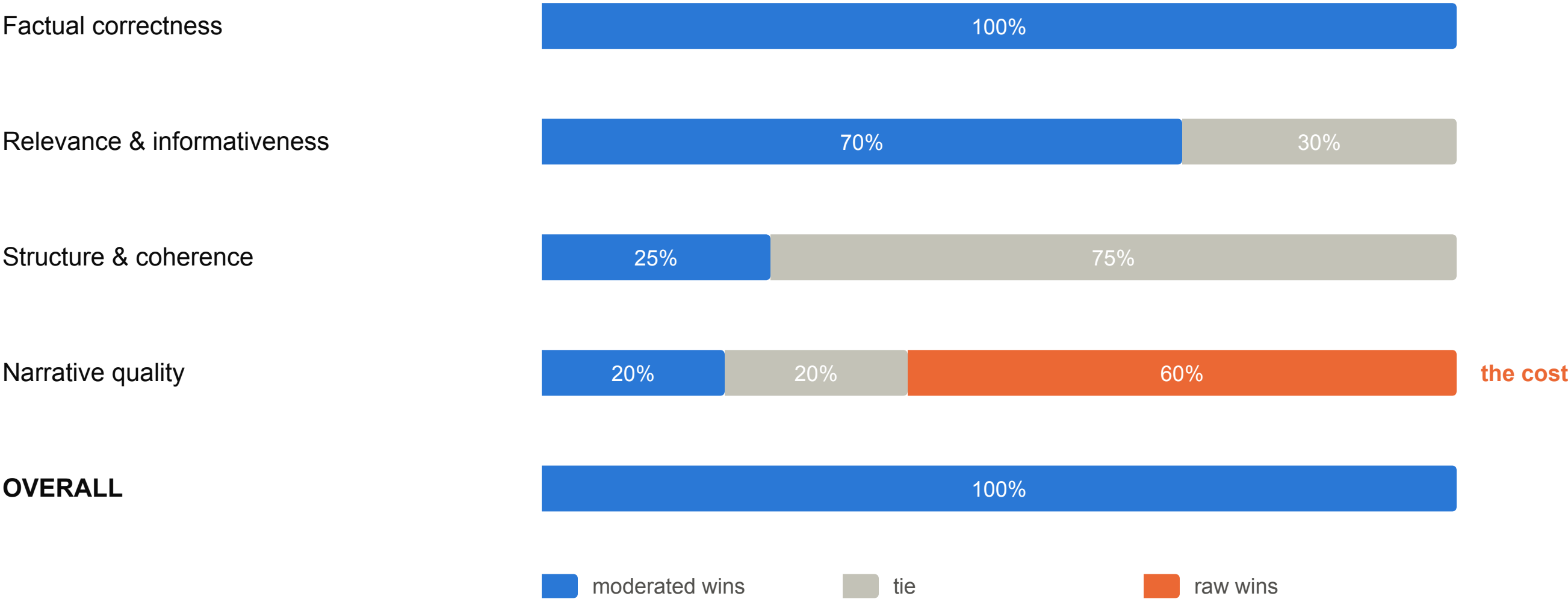
Raw alarmism across the qwen3.5 ladder: **4.8** at 2B, **4.6** at 4B, **4.5** at 9B, **4.3** at 27B. Scaling by more than 13× changes nothing.

The one calm generator in the set is **llama3.1:8b at 3.98**. Our earlier reading, that smaller generators write hotter copy, was a family effect misread as a scale one.

Same series, same seed, same moderator throughout

The moderated story wins on everything except readability

20 blinded pairs, position randomised, Claude Opus deciding each criterion.



The judge is never told which story was moderated, and which one is shown first is decided by a coin flip per pair. Narrative quality is the one criterion the raw story wins, and it wins it three times as often: this system buys calibration and accuracy by spending engagement, and a deck that hides that row is selling rather than reporting.

The honest slide

- **Narrative quality is the price.** The unmoderated story wins it three times as often. We buy calibration and accuracy by spending engagement.
- **n = 1 outside two cells.** Every other combination is a single run, and we have shown what that can do to a conclusion.
- **One judge.** We measured how much it agrees with itself, not how much a different judge would agree with it.
- **Seven datasets are about three.** Five are WHO series of the same shape and two are nested.

IN ONE LINE

Tone moderation reaches the human level, and the measurement holds up.

Alarmism 3.74 → 2.09 against a human 2.00, 100% on factual correctness, judge ICC 0.99. The costs are narrative quality and a sample size we are still honest about.